

Global and Local Projection-Based Reduced-Order Models

2D Burgers and Hypersonic Vehicle Campaigns

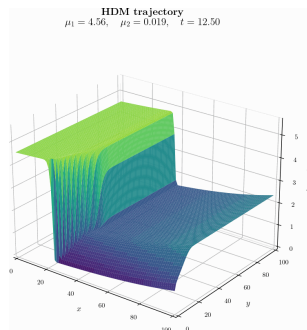
Sebastian Ares de Parga¹, Radek Tezaur¹, Charbel Farhat^{1,2}

¹Department of Aeronautics and Astronautics

²Institute for Computational and Mathematical Engineering
Stanford University, Stanford, CA 94305, USA

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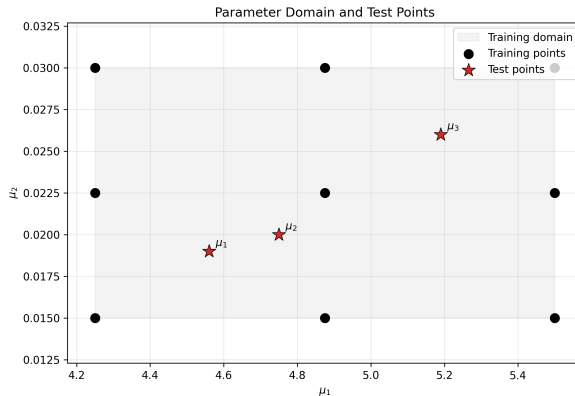
2D Burgers Campaign



Global and local HPRM validation

Linear POD bases, quadratic manifolds, and GPR manifolds

Parameter Domain and Test Points



Online Performance (250×250): Avg/Max Error over 3 Test Points

Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,u}$ (%)	max $\mathbb{RE}_{2,u}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.517	–
HPROM	96	–	4 241	1.030	1.096	0.366	34.2
HQPROM	39	–	3 460	0.818	0.861	0.641	19.5
HPROM-GPR	20	131	1 799	0.666	0.785	0.236	53.1
Local HPROM ($N_c = 3$)	35–48	–	3 405	0.897	0.933	0.154	81.4
Local HQPROM ($N_c = 3$)	21–27	–	2 152	0.888	1.025	0.215	58.3
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.714	0.786	0.113	110.5
Local HPROM ($N_c = 5$)	23–31	–	2 384	0.915	0.943	0.095	132.0
Local HQPROM ($N_c = 5$)	15–20	–	1 593	0.760	0.878	0.128	97.9
Local HPROM-GPR ($N_c = 5$)	7	41–68	631	0.673	0.880	0.094	133.2

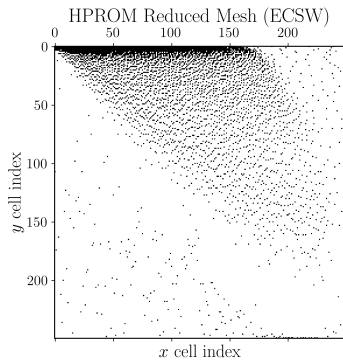
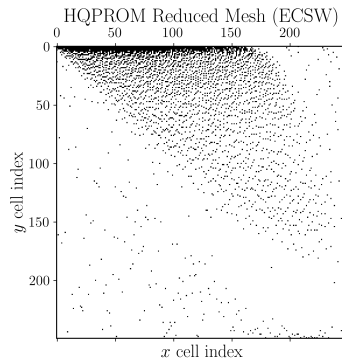
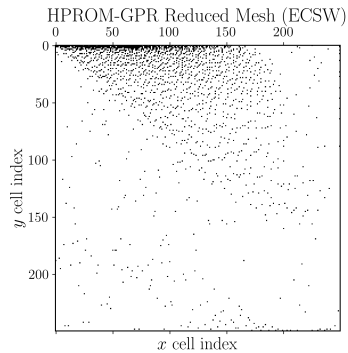
Speedup is computed with respect to the HDM runtime, $t_{\text{HDM}} = 751.043$ s, averaged over the three test points.

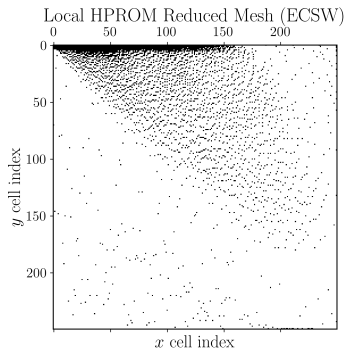
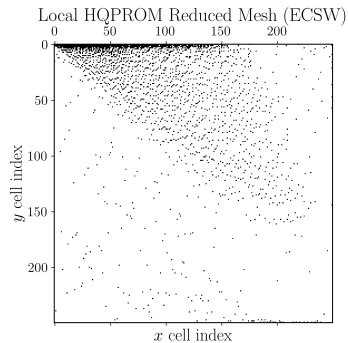
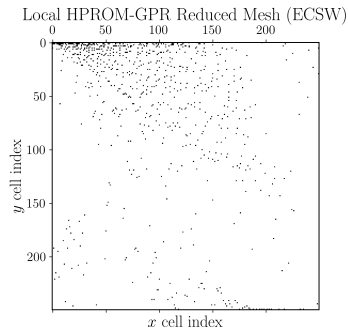
Online timings were measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

N_e denotes the number of sampled ECSW elements ($N_e = 62,500$ for HDM/full mesh).

All models are reported at a matched accuracy target of $1\% \pm 0.2\%$ in max $\mathbb{RE}_{2,u}$: for each N_c the accuracy knob is tuned into that window (pod.to1 for Local HPROM, ζ_{qua} for Local HQPROM, n for Local HPROM-GPR), so differences in n , N_e and speedup are read at constant accuracy.

Global Reduced Meshes (ECSW)

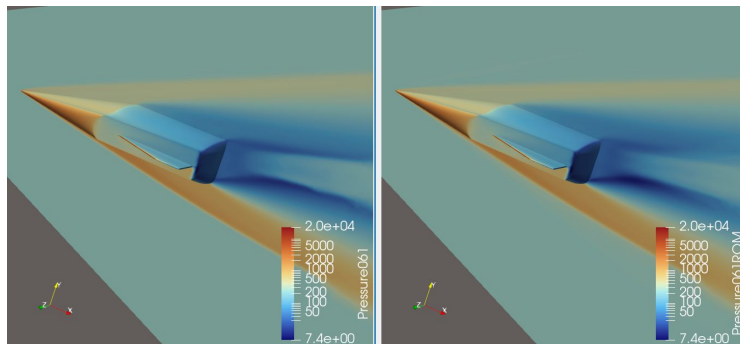
HPROM ($N_e = 4,241$)HQPROM ($N_e = 3,460$)HPROM-GPR ($N_e = 1,799$)

Local Reduced Meshes (ECSW), $N_c = 3$ Local HPROM ($N_e = 3,405$)Local HQPROM ($N_e = 2,152$)Local HPROM-GPR ($N_e = 811$)

Burgers Conclusions

- ▶ At matched accuracy, every local ROM is faster than its global counterpart: $34.2\times \rightarrow 132.0\times$ (HPROM), $19.5\times \rightarrow 97.9\times$ (HQPROM), $53.1\times \rightarrow 133.2\times$ (HPROM-GPR).
- ▶ Within each family the speedup grows monotonically with N_c , so clustering is the dominant lever rather than the choice of local manifold.
- ▶ When the nonlinear manifold does not compress enough, as for the quadratic, the speedup reached at a given error need not beat the linear model. This is problem-dependent, so the three families are alternatives rather than a ranking.
- ▶ Clustering and nonlinear compression are not additive. A linear basis always gains from more clusters, whereas HPROM-GPR already carries an intrinsic compression that clustering should support rather than replace: at $N_c = 5$ the local linear HPROM has caught up with it ($132.0\times$ versus $133.2\times$), and raising N_c further would make HPROM-GPR more expensive, since every nonlinear step also pays for the manifold tangent $\Psi = \mathbf{V} + \bar{\mathbf{V}} \partial \mathcal{N}_{\text{GPR}} / \partial \mathbf{q}$, the closure evaluation, and cluster selection and transfer.

Hypersonic Vehicle Campaign

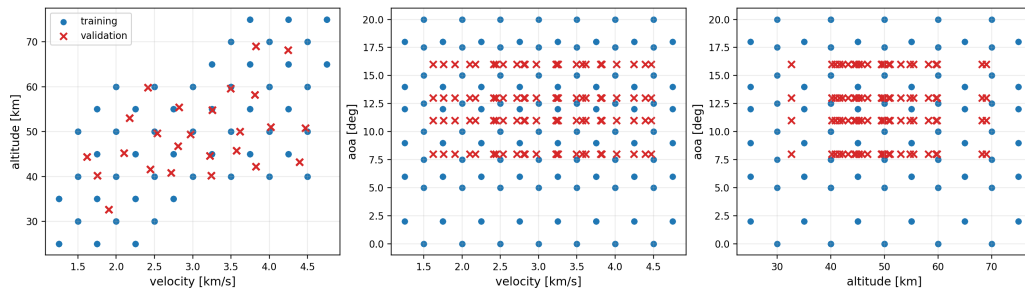


Global and local HPRM validation

Linear POD bases and RBF manifolds

Training and Validation Space

363 HDM training cases; 100 independent HDM validation cases; four local clusters with 20% overlap; ECSW tolerance 10^{-3} .



HGV2 ROM Configurations

Model	Reduction	Closure	N_c
HPROM	one POD basis	–	1
HPROM–RBF	$n = 8$ primary coordinates	one RBF map	1
Local HPROM	one POD basis per cluster	–	4
Local HPROM–RBF	$n = 8$ primary coordinates	one RBF map per cluster	4

Every POD basis and RBF map is trained from the same 363 HDM parameter cases. Each online state is assigned to its nearest cluster. ECSW is an independent offline construction step.

HGV2 Online Performance and Validation Accuracy

Computational model	n (N for HDM)	$\bar{n}$	C_D (%)	C_L (%)	C_M (%)	Online time (s)	Speedup
HDM	5 674 883	–	–	–	–	<i>pending</i>	–
HPROM	362	–	0.975	0.554	6.620	24.03	–
HPROM–RBF	8	354	1.388	1.335	7.004	1.15	21.0
Local HPROM ($N_c = 4$)	95–168	–	1.101	0.444	6.242	6.80	3.5
Local HPROM–RBF ($N_c = 4$)	8	117–140	1.978	2.183	7.117	0.62	38.9

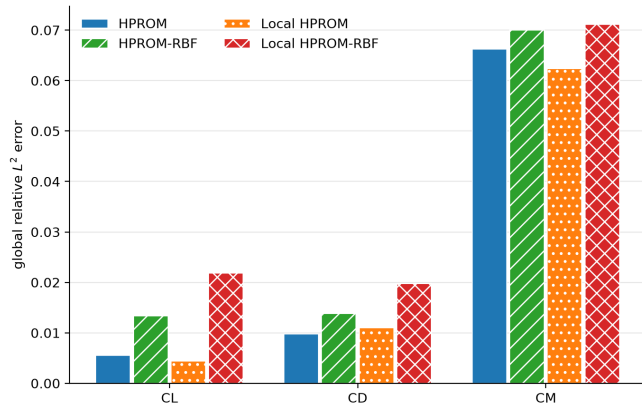
Coefficient errors are global relative L^2 values over 100 independent validation cases.

Speedup is computed with respect to the HPROM online time, so HPROM itself carries no entry. The HDM runtime is pending.

Timings are means over 10 validation cases, measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

For the HDM, N is the number of mesh nodes. For HPROM, n is the full POD rank. For HPROM–RBF, n and $\bar{n}$ are the primary and secondary dimensions; local $\bar{n}$ is the range across the four cluster bases.

HGV2 Validation Error Comparison



HGV2 ECSW Offline Cost

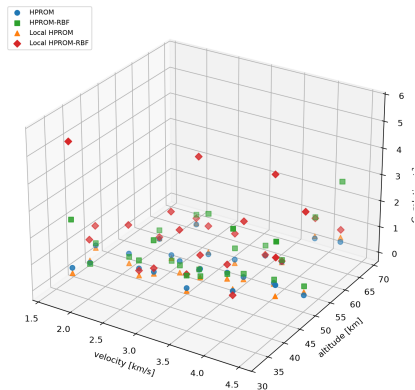
Computational model	Training rows	Sample nodes	ECSW (h)
HPROM	27 000	5 089	30.04
HPROM–RBF	5 808	1 715	0.43
Local HPROM ($N_c = 4$)	27 000	3 484	9.95
Local HPROM–RBF ($N_c = 4$)	1 920	1 113	0.38

ECSW was run on Sherlock using 20 nodes, 480 MPI ranks, and 480 cores. Times are maximum-rank wall times.

Sample nodes is the size of the reduced mesh. How the training rows are assembled is detailed in the appendix.

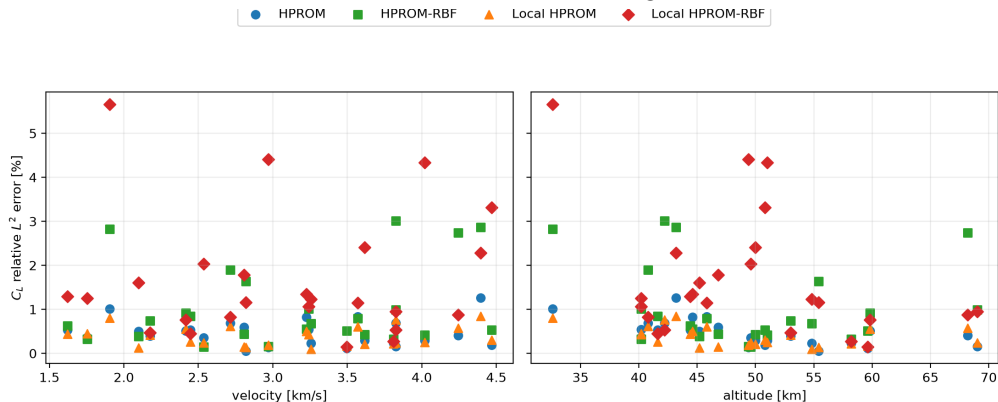
HGV2 Three-Dimensional C_L Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_L error over the four sampled angles of attack.



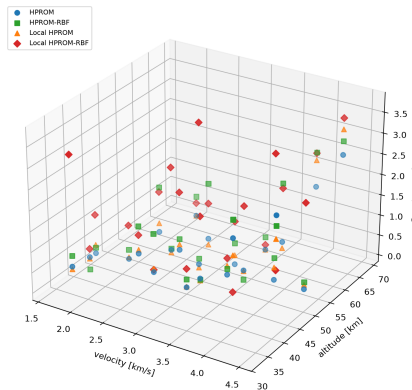
HGV2 Two-Dimensional C_L Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



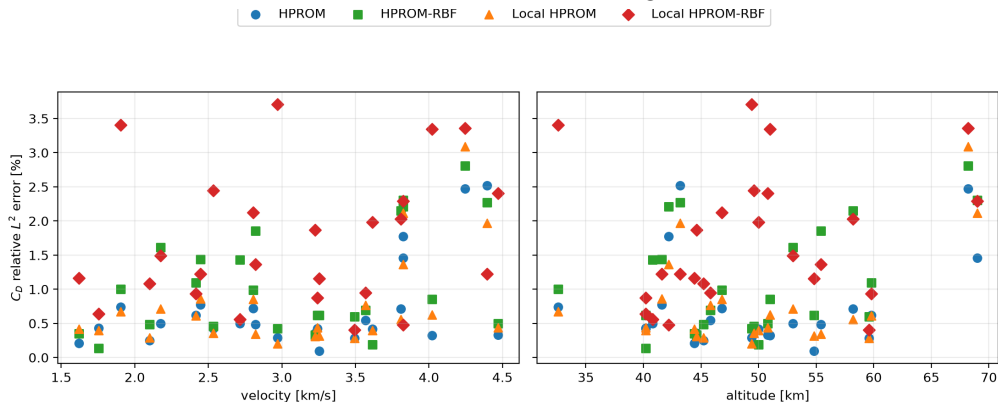
HGV2 Three-Dimensional C_D Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_D error over the four sampled angles of attack.



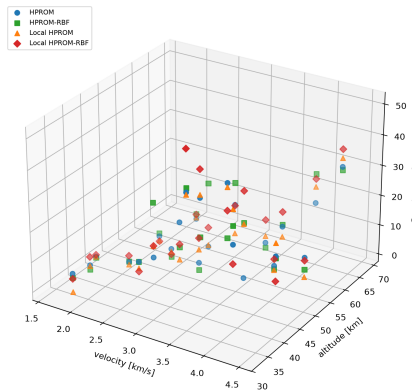
HGV2 Two-Dimensional C_D Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



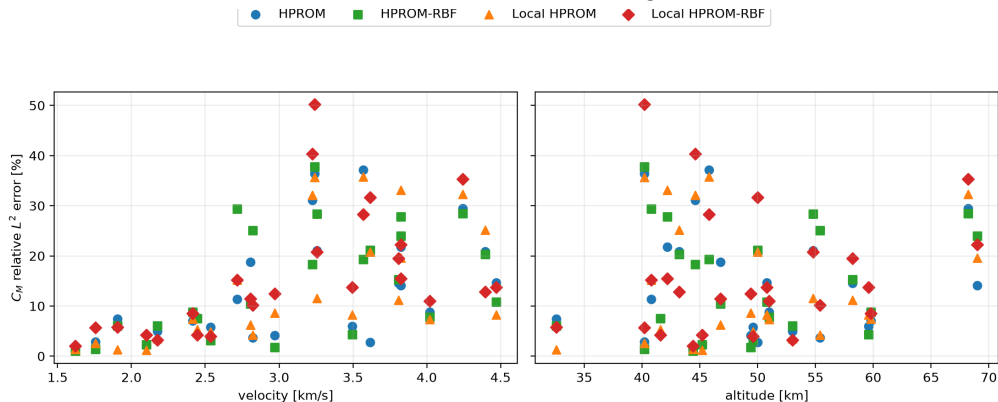
HGV2 Three-Dimensional C_M Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_M error over the four sampled angles of attack.



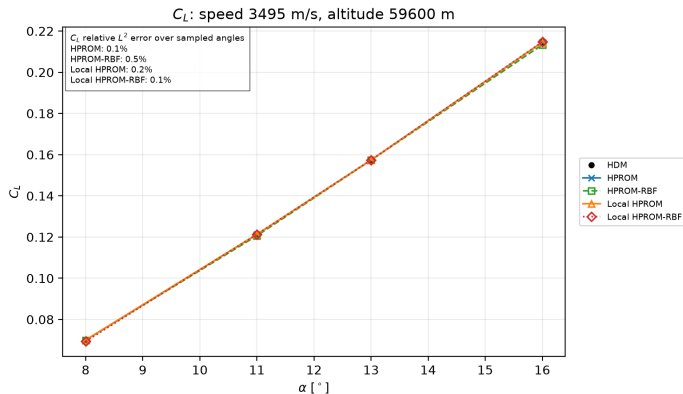
HGV2 Two-Dimensional C_M Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



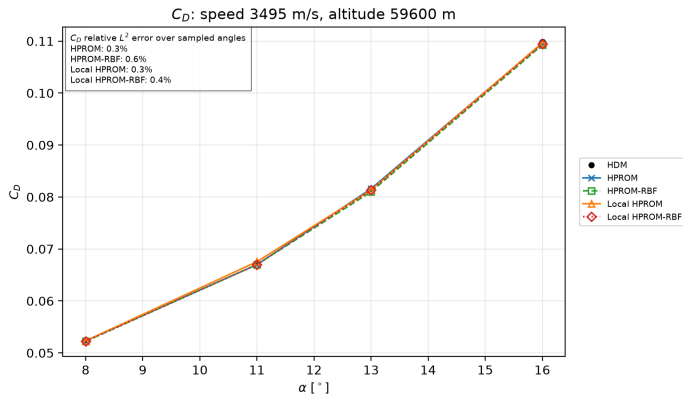
HGV2 Representative Median Pair: C_L

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



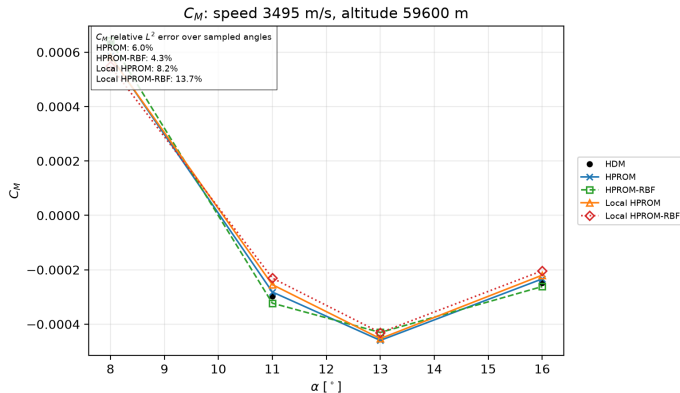
HGV2 Representative Median Pair: C_D

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



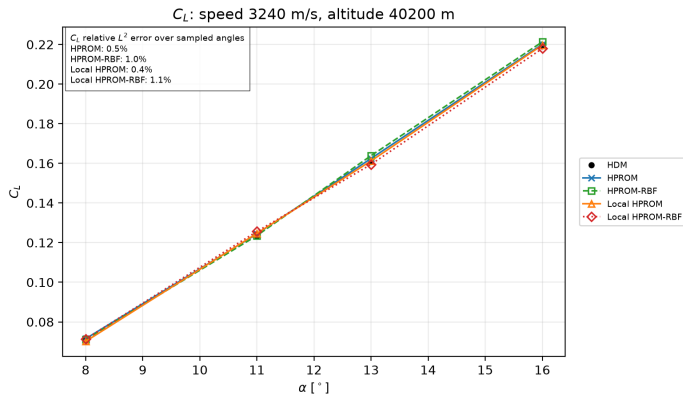
HGV2 Representative Median Pair: C_M

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



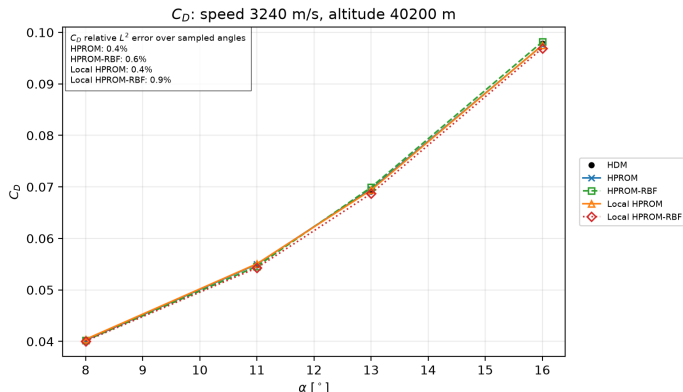
HGV2 Representative Challenging Pair: C_L

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .



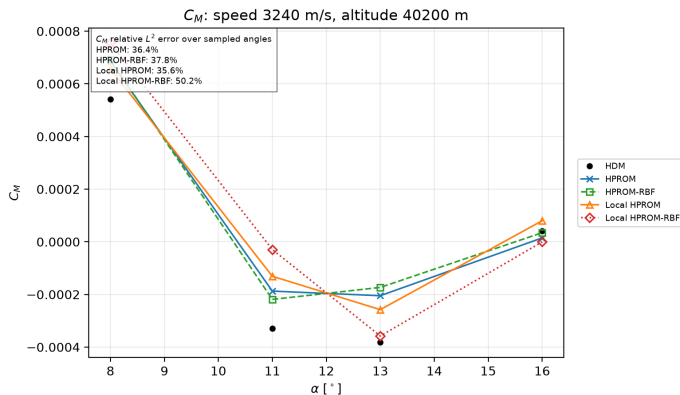
HGV2 Representative Challenging Pair: C_D

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .



HGV2 Representative Challenging Pair: C_M

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across C_L , C_D , and C_M .



HGV2 Conclusions

- ▶ The four models behave similarly in accuracy: C_M falls within 6.2–7.1% for all of them, and C_D and C_L within 1–2%. The linear models are the most accurate and the RBF models the fastest, so they are offered as alternatives rather than as a ranking.
- ▶ HPRM–RBF already supplies the compression: $n = 8$ against a full POD rank of 362. Adding four clusters does not reduce n further, but it accelerates the model twice over — the reduced mesh shrinks from 1,715 to 1,113 sample nodes, and each RBF evaluation is cheaper because the per-cluster maps are smaller ($\bar{n}$ drops from 354 to 117–140).
- ▶ The offline cost separates the two families by almost two orders of magnitude: 30.04 h against 0.43 h for the global models, and 9.95 h against 0.38 h for the local ones.

HQPRM and Local HQPRM are not included in this campaign.

Supplementary results and implementation details

Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^* = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta \mathbf{q}$ is obtained from

$$(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}) \Delta \mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

Global Linear HPRM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Global HQPROM

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Global HPROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg \min_{\ell=1,\dots,N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2\mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$ no transfer, $k^+ \neq k^- \Rightarrow$ solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

Appendix: Burgers Paper Campaign Setup

Two-dimensional parametric inviscid Burgers problem

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}.$$

$$\mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}, \quad u_x(0, y, t; \boldsymbol{\mu}) = \mu_1.$$

 μ_1 : inflow value μ_2 : exponential source variation $\boldsymbol{\mu} \in [4.25, 5.50] \times [0.015, 0.030]$

Test points:

$$\boldsymbol{\mu}_1 = (4.56, 0.019), \quad \boldsymbol{\mu}_2 = (4.75, 0.020), \quad \boldsymbol{\mu}_3 = (5.19, 0.026).$$

HDM: $\Omega = [0, 100]^2$, 250×250
finite-volume cells.

Methods included:

Global HPROM

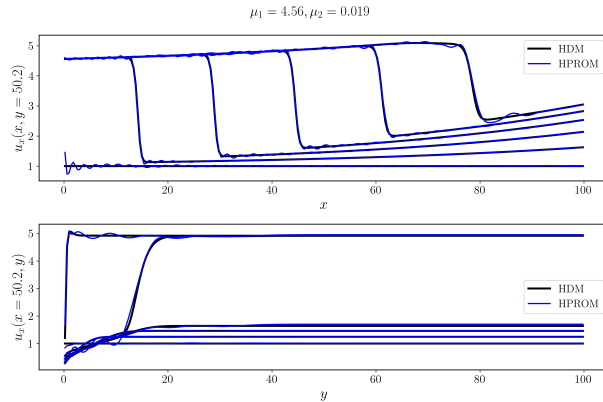
Local HPROM

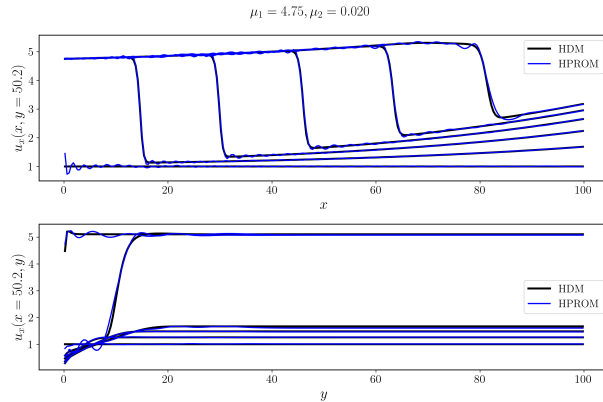
Global HQPROM

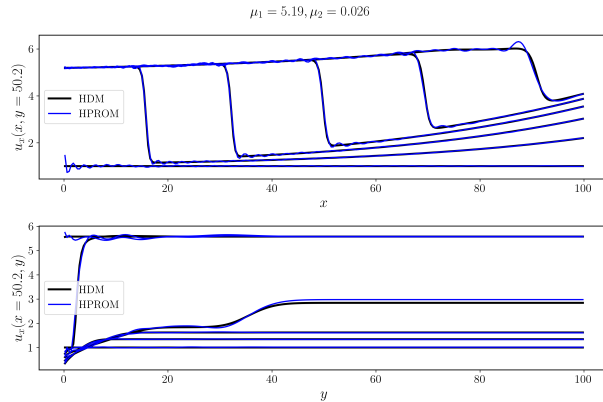
Local HQPROM

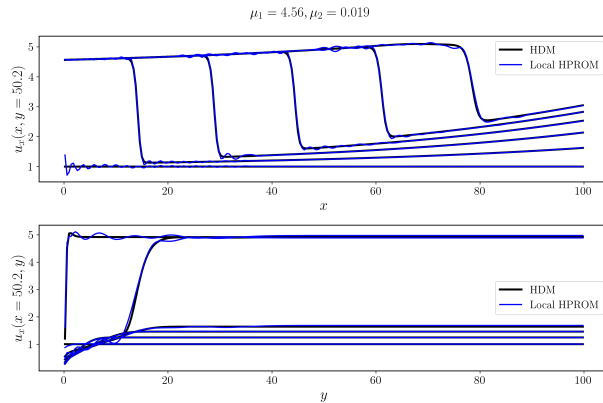
Global HPROM-GPR

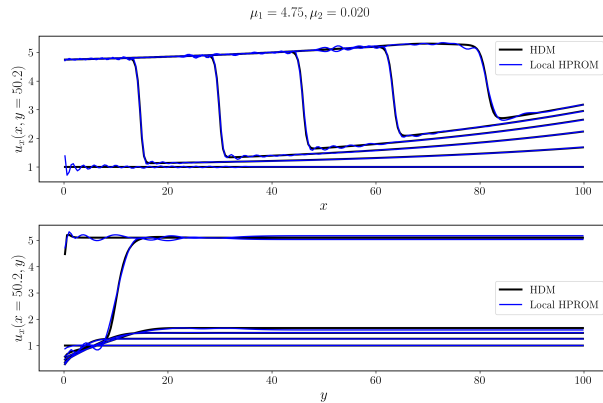
Local HPROM-GPR

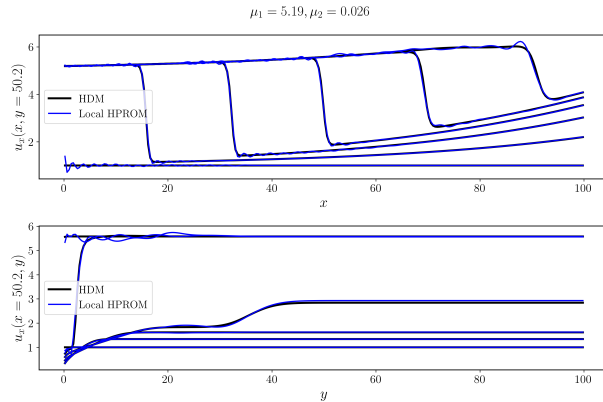
HPROM vs HDM: $\mu = (4.56, 0.019)$ 

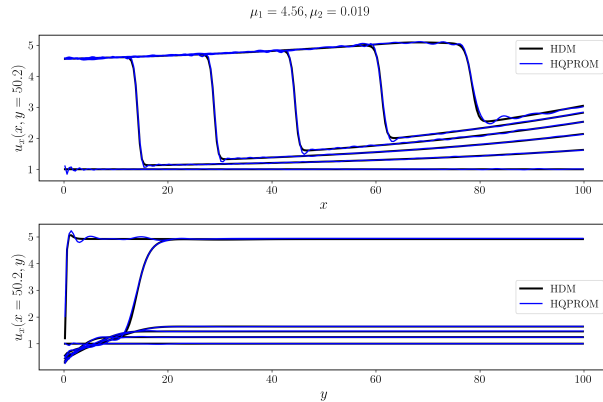
HPROM vs HDM: $\mu = (4.75, 0.020)$ 

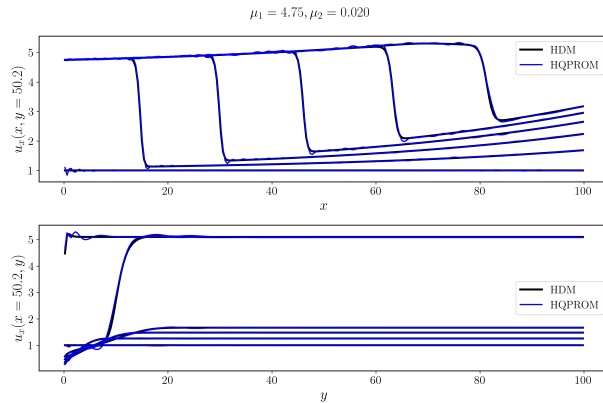
HPROM vs HDM: $\mu = (5.19, 0.026)$ 

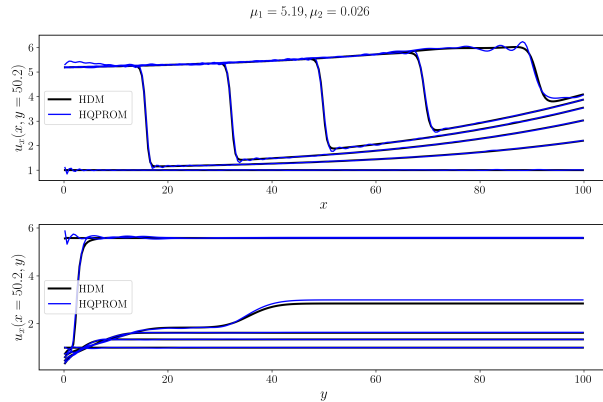
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

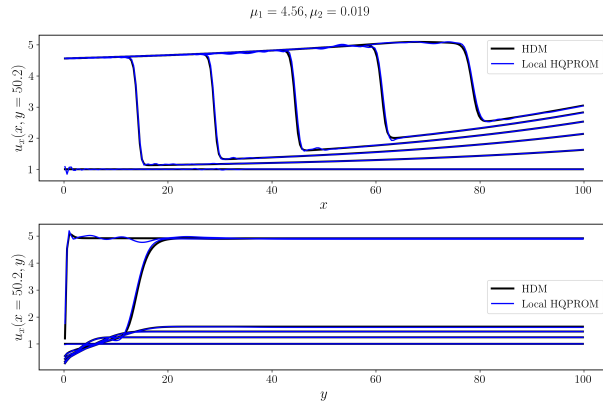
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

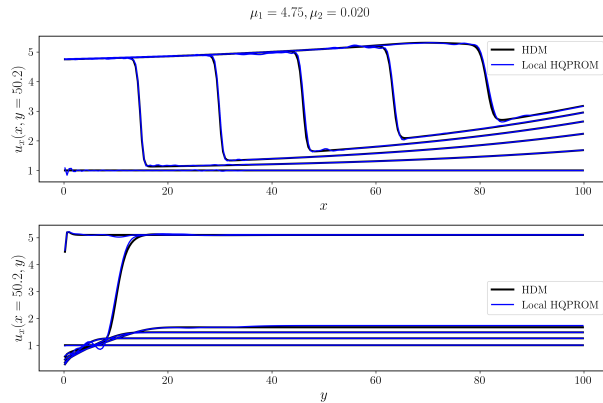
Local HPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

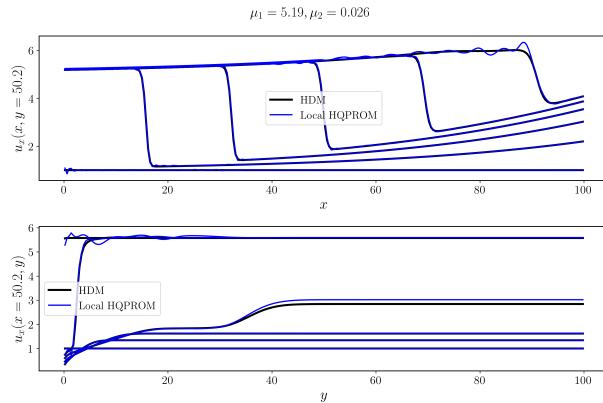
HQPROM vs HDM: $\mu = (4.56, 0.019)$ 

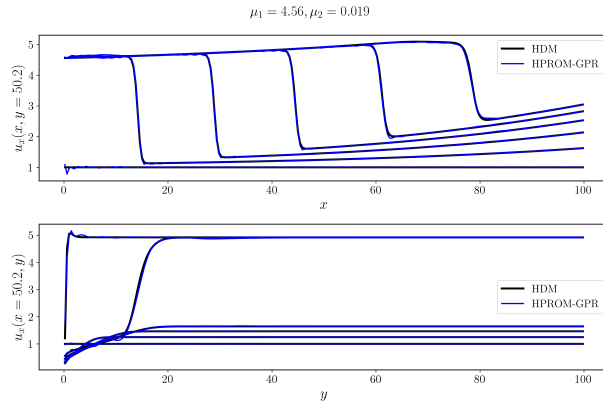
HQPROM vs HDM: $\mu = (4.75, 0.020)$ 

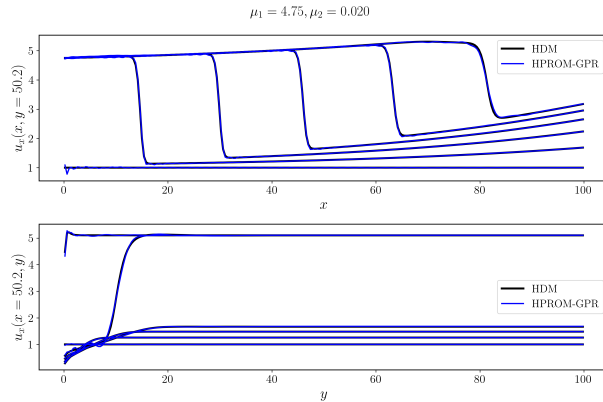
HQPROM vs HDM: $\mu = (5.19, 0.026)$ 

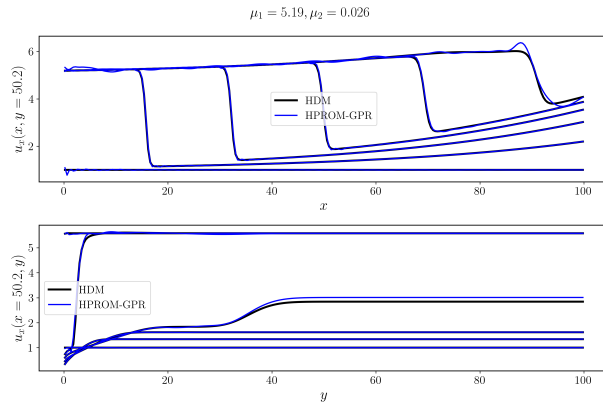
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

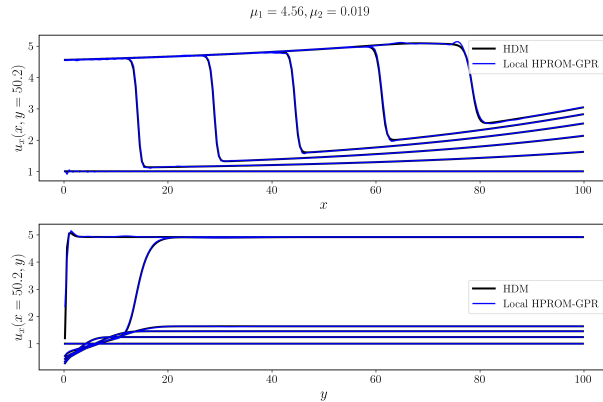
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

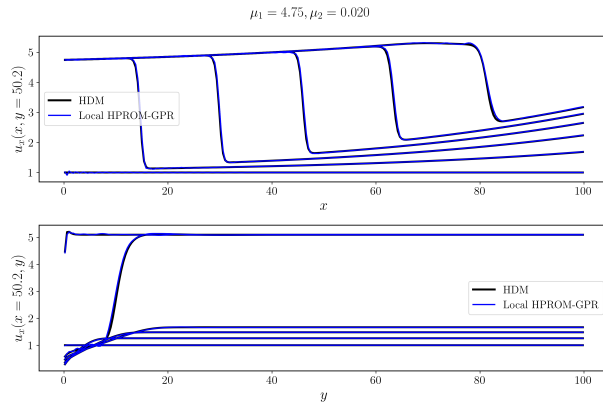
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

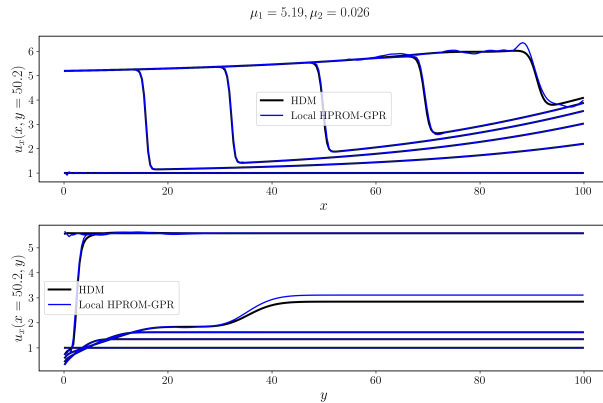
HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$ 

HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$ 

HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

Appendix: ECSW Training Data

For `DataType = Jacobian`, each selected parameter solution assigned to cluster k supplies m_k^2 candidate ECSW rows, where $m_k = \min(r_k, \text{MaxNumResidualComponents})$ and `MaxNumResidualComponents` = 30. With row stride s , CM reads every s -th row of its `training_rows` sequence.

Model	ECSW cases	Row stride	m_k	Rows read by CM
Global linear HPROM	30	1	$\min(362, 30) = 30$	$30 \times 30^2 = 27,000$
Local linear HPROM, $N_c = 4$	30	1	$\min(r_k, 30) = 30$	$30 \times 30^2 = 27,000$
Global HPROM-RBF, $n = 8$ (ablation)	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$
Global HPROM-RBF, $n = 8$ (final)	363	4	$\min(8, 30) = 8$	$363 \times 8^2 / 4 = 5,808$
Local HPROM-RBF, $N_c = 4$, $n = 8$	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$

Interpretation

An ECSW snapshot is a Jacobian-training row, not a physical-time state. The RBF primary dimension $n = 8$ —not $\bar{n}$ —sets the 8×8 count. A row stride of 4 reads rows 0, 4, 8, . . . within every selected parameter case; it does not subsample the 363 parameter cases and does not refer to online Newton iterations.

Appendix: HGV2 Reduced-Mesh Construction

Model	L0	L1	L2	L3	Nodes	Elms.	N_{-L2}
Global linear HPROM	5,089	36,971	166,878	60,623	269,561	563,301	92,683
Local linear HPROM, 4 clusters	3,484	28,145	169,798	47,183	248,610	470,649	78,812
Global HPROM-RBF, $n = 8$	1,715	15,809	173,072	28,648	219,244	350,921	46,172
Local HPROM-RBF, 4 clusters, $n = 8$	1,113	10,516	175,813	20,602	208,044	298,219	32,231

L0: selected ECSW sample nodes.

L1: nodes added by edge connectivity for linear reconstruction.

L2: lift-surface nodes used only for C_L , C_D , and C_M postprocessing.

L3: nodes added by element connectivity for state reconstruction.

N_{-L2} excludes the postprocessing-only lift-surface nodes.